Deep Learning-Based Face Morph Attack Detection: A Comparative Study of Transfer Learning Approaches

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***Abstract*—Face morph attacks constitute a considerable threat to modern biometric systems used in passport verification and border security applications due to their ability to blend facial features of several people into one picture. Such face morph attacks have the potential to trick both humans and machine-based face recognition systems.**

In this research work, four different deep learning methods for face morph attack detection were implemented on the AMSL dataset with 2,379 facial pictures (including morphs). All experiments were done on similar terms, with four tested architectures, namely a baseline CNN approach and three transfer learning algorithms. Accuracy, Precision, Recall, F1-Score, and ROC-AUC values were used to evaluate all four algorithms.

Results obtained through the evaluation of various architectures showed the advantage of transfer learning compared to the regular CNN architecture. Specifically, among all the tested architectures, the most successful was InceptionV3, which yielded the following metrics: accuracy of 97.44%, precision of 1.000, recall of 0.7069, F1-score of 0.8283, and ROC-AUC of 0.9916.

The conclusion drawn from these experiments is that transfer learning approaches significantly outperform CNN in detecting morphs by extracting more subtle characteristics.

Keywords— Face Morph Attack Detection, Face Morphing, Deep Learning, Transfer Learning, InceptionV3, Biometric Security, Facial Recognition, AMSL Dataset

I. Introduction

A. Background and Motivation

The biometric authentication has become a crucial part of present identity verification solutions. Various biometric modes have been adopted for identity verification. However, face recognition, due to its unobtrusive nature, ease of acquisition and capability of working in digital world, is one of the widely adopted biometric modes. Face recognition systems have already been integrated into electronic passports, border control, CCTV, mobiles and other access control systems. Recent progress in AI and computer vision field has greatly enhance the accuracy and robustness of facial recognition systems. Deep learning based face recognition systems can extract more discriminate features than traditional methods so it can reliably identify and verify faces under varying conditions. Thus, facial biometrics has become a desirable solution for identity verification in both civilian and government fields. However, increased dependence on facial recognition technology brings new security concerns. The wide accessibility of face manipulation and face synthesis tools is used by attackers to create realistic face forgery, which can effectively fool biometric identification systems. Thus, securing the credibility of facial biometrics is becoming an important research issue. It encourages the development of sophisticated detection methods for face manipulation to prevent biometric authentication systems from attacks.

B. Face Morphing Attacks and Challenges

Face morphing attacks constitute an important challenge to the modern biometric authentication systems. The image obtained by merging the facial characteristics of two or more faces into a synthetic image that can be verified against more than one identities is defined as morph image. This type of attacks is of great concern especially to the electronic passport and border control applications where the verification against the morphed image can result in an unreliable identity confirmation.

C. Existing Solutions and Research Gap

In order to combat morphing attacks, researchers have proposed handcrafted features-based approaches, texture-based approaches and deep learning-based approaches. Some of the morphing artifacts could be detected using the traditional methods but their efficiency can not guarantee while being tested on the variety of databases and the novel morphing generation methods. Recently deep learning approaches using the CNN structures and the transfer learning based methods are the promising directions for the feature learning. They learn the discrimination representations for face morph attack detection.

In addition, some Vision Transformer based approaches [15] and advanced optimization-based frameworks [16] provide satisfactory results but they might be too complex for real biometric systems. While a fair comparison across widely-used transfer learning-based architectures has not been yet performed, this work aims to make a comparative study using CNN, MobileNetV2, ResNet50 and InceptionV3 on AMSL Face Morph Image Dataset in order to obtain an appropriate and dependable system to detect face morph attacks.

D. Research Objectives and Contributions

This work aims to explore the effectiveness of deep learning approaches for detecting face morphing attacks in biometric authentication systems. In pursuit of this aim, a robust experimental setup was established utilizing the AMSL Face Morph Image Dataset to train and test a variety of deep learning architectures in a unified manner, so as to measure how suitable each is to detect morph attacks. In this paper, four deep learning models (CNN, MobileNetV2, ResNet50, and InceptionV3) are implemented and trained on binary classification task of real and morphed face images. All standard image preprocessing techniques, data augmentation and evaluation criteria were applied uniformly on all architectures to ensure their comparative analysis in a fair manner. Accuracy, Precision, Recall, F1-Score and ROC-AUC were employed to measure their performance. The key contributions of this paper include

- Development of a complete deep learning-based face morph attack detection framework using the AMSL Face Morph Image Dataset.
- Comparative evaluation of CNN, MobileNetV2, ResNet50, and InceptionV3 architectures under identical experimental conditions.
- Analysis of model performance using multiple classification and discrimination metrics.
- Identification of the most effective architecture for robust face morph attack detection.
- Provision of insights that may support the development of secure and reliable biometric authentication systems.

II. Related Work

A. Traditional Morph Detection Methods

Early attempts at face morph attack detection focused on traditional handcrafted feature extraction and conventional machine learning algorithms. These techniques concentrated on detecting anomalies generated during the morphing process through analysis of image texture variations, quality degradation, and frequency characteristics of the images. Traditional features, such as LBP, BSIF, HOG, and SIFT, have often been employed in studies. Texture-based methods were applied to reveal pixel-level differences between real and morphed images in detail. Frequency-domain based methods, employing Discrete Cosine Transform (DCT) and Fourier analysis, were investigated to uncover the morphing artifacts that could not be detected visually in the spatial domain. Furthermore, the effectiveness of image quality assessment techniques was explored to detect inconsistencies resulted from blending and post-processing in morph generation. While traditional methods could achieve respectable accuracy on constrained test datasets, they suffered from limited robustness and generalization capability, mainly due to factors such as lighting changes, pose variations, expression variability, inconsistent image quality, and differing morphing strategies. Moreover, handcrafted features are often highly sensitive to those factors and typically need domain-specific knowledge and time-consuming engineering. These weaknesses drove the research toward the development of deep learning methods for morph attack detection, allowing algorithms to learn meaningful representations directly from the image data.

B. Deep Learning-Based Morph Detection Methods

The rise of deep learning approaches has revolutionized face morph attack detection. Traditional methods relies on hand-engineered feature extraction from images whereas deep learning networks can directly learn hierarchal features from input images. These hierarchical features are able to extract subtle morphological inconsistency which can barely be detected by image processing methods.

The main components of state-of-the-art deep learning based morph attack detection models are CNNs due to their outstanding feature extraction and classification performance. Some prior researches demonstrated their ability to identify real faces from morphs by learning discriminative spatial patterns. With the availability of more dataset and resources, deep learning has achieved better results than traditional feature-based methods on various benchmark datasets. Transfer learning methods which use pre-trained deep networks like MobileNet, ResNet, DenseNet, EfficientNet, Xception and Inception networks have received increasing attention due to their effectiveness and faster training times in biometric security tasks, especially when there is less data available to train from scratch.

C. Research Gap and Motivation

Although substantial work has been made on the detection of face morph attack, a number of research gaps are yet to be filled. Few research works evaluate various deep learning architectures or detection methods for the sake of comparing their performances on an equal basis. Besides, various discrepancies in preprocessing techniques, dataset used, training protocol, and evaluation measures, sometimes result in differing results across many literatures. Therefore, to what extent are popular transfer learning architectures efficient for face morph attack detection is not thoroughly evaluated.

In this study, four deep learning architectures (CNN, MobileNetV2, ResNet50, InceptionV3) are conducted on AMSL Face Morph Image Dataset through the same procedure of preprocessing, augmentation, training and testing. We want to determine the most efficient architecture for secure face morph attack detection. And benchmark experiments. The purpose of this research aims to be an assistant work for the biometrics security system design.



Fig. 1. Representative samples from the AMSL Face Morph Dataset.

TABLE I
CHARACTERISTICS OF THE AMSL FACE MORPH DATASET USED IN THIS STUDY

Parameter	Value
Dataset Name	AMSL Face Morph Image Dataset
Source Dataset	Face Research Lab London Set
Application Domain	Face Morph Attack Detection
Number of Classes	2 (Genuine, Morph)
Genuine Images	204
Morph Images	2175
Total Images	2379
Class Distribution	8.57% : 91.43%
Morph Generation Method	Combined Morph ($\alpha = 0.5$)
Image Resolution	531 $\times$ 413 pixels
Image Format	JPEG
Input Size	224 $\times$ 224 $\times$ 3

III. PROPOSED METHODOLOGY

A. Dataset Description

For this research, all the experiments were performed using the AMSL Face Morph Image Dataset [10], which is a publicly available benchmark dataset, and has been used to test and develop the various face morph attack detection techniques. It comprises of both genuine faces and attack images (morph images), that were generated using controlled face morphing technique, which morphs faces between two subjects while following the standard of biometric document.

AMSL dataset was generated from Face Research Lab London Set [17], and was introduced as a set of images useful for testing biometric system security against morphing attacks. The images from the set were processed with the ICAO portrait quality standards (alignment, cropping, scaling, and compression), therefore resembles the actual passport/identity document situation, that makes it suitable for research in morph attack detection. The AMSL dataset used in this research has 204 genuine facial images, 2175 morph images (attack sample) that sum up to 2379 images. The images were organized in 2 binary classification groups: *Real* and *Morph* as listed in Table I.

The genuine and morph samples provided by the AMSL dataset creates a suitable test bed for evaluating the robustness of deep learning based morph attack detection algorithms, as it is often found that the number of attack images in test sets is much larger than genuine enrolled samples in real world scenarios, therefore enabling testing for the case of class imbalance.

B. Data Preprocessing and Augmentation

A thorough and effective data preprocessing pipeline helps to achieve faster model convergence, prevents overfitting of models and maintains consistency of features extracted among the deep learning models. Before feeding images into model for training, all images were categorized into two groups as *Real* and *Morph* which represent the original face image and morphed image respectively. Images were then preprocessed for experiments on binary classification.

To make the data consistent among all the trained models, images were resized into fixed resolution of 224×224 pixels. All pixel values were normalized by scaling intensities of each images to be in between 0 and 1 which stabilize model training numerically and used for custom CNN model as well as transfer learning models.

Augmentation was performed on random rotation(10), zoom transformation(0.1) and horizontal flip. These augmentation processes generated more data variations artificially.

After preprocessing, the images were split into training set (70%), validation set (15%) and testing set (15%) for model training, tuning hyper parameters and evaluation respectively.

C. Proposed Detection Pipeline

This document describes a proposed framework for face morph attack detection that leverages a typical deep learning workflow, encompassing data preparation, data preprocessing, model training, model validation, and classification. The core idea of this workflow is to automatically identify genuine images from morphed faces with satisfactory classification accuracy and robustness. First, images of real faces from AMSL dataset were collected and then categorized into morph and genuine class. Afterwards, preprocessing techniques like resizing, normalization and augmentation have been applied to images so as to provide consistent input and promote deep learning model learn more discriminative facial representations. Once the image has been preprocessed, training set, validation set and testing set were obtained. Training set was used to train the parameters of the model and validation set used to supervise training and prevent over fitting, and the testing set used for final evaluating. Then four deep learning models named CNN, MobileNetV2[18], ResNet50[19] and InceptionV3[20] are trained under same conditions and tested using various metrics including Accuracy, Precision, Recall, F1-Score, ROC-AUC. Among the four models, the best one was selected as the model to detect face morph attacks finally. Figure shows the overall procedure in this work to detect face morph attacks.

Fig. 2.

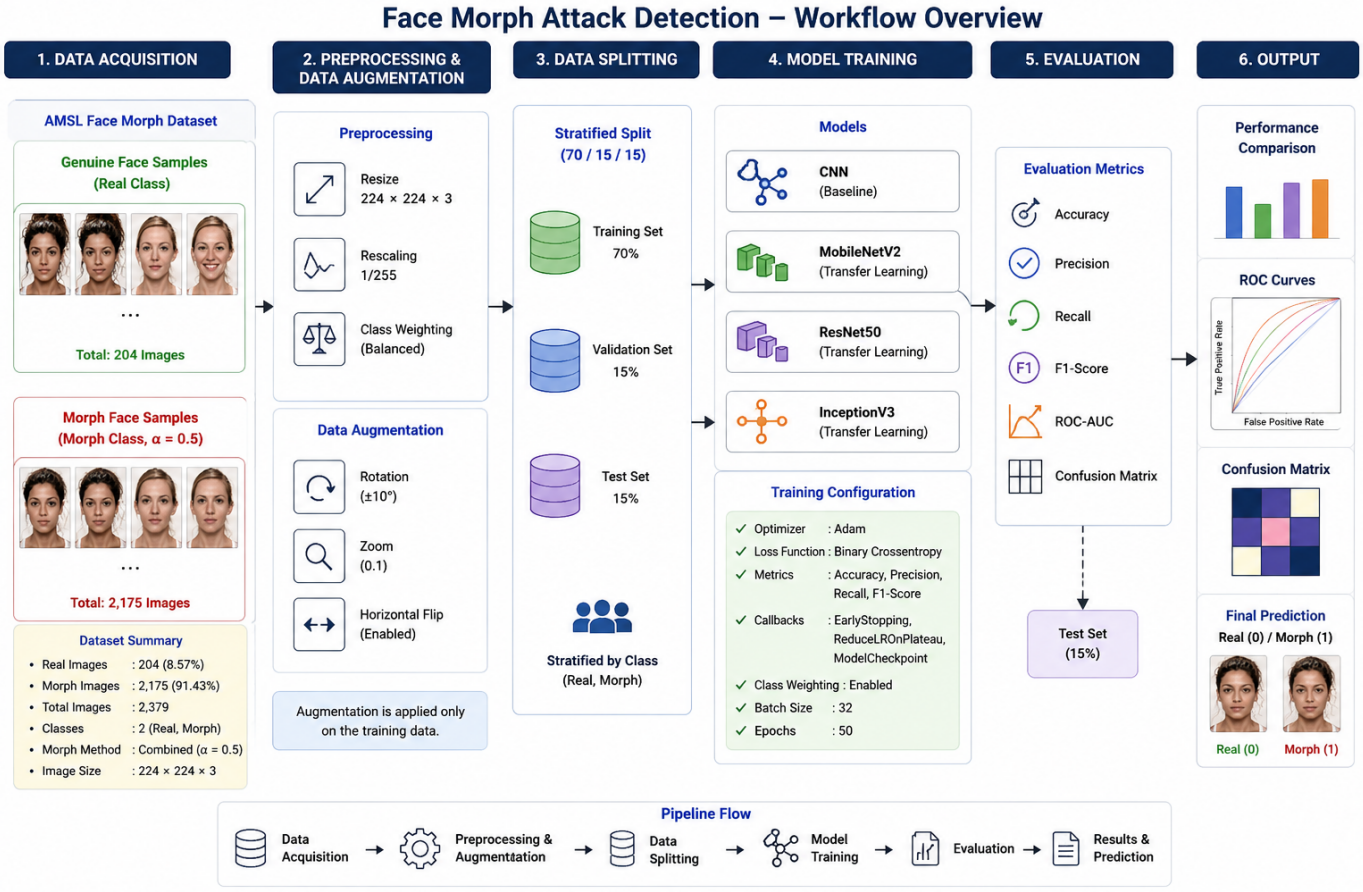


Fig. 2. The overall workflow adopted in this study for face morph attack detection is illustrated in Fig. 2.

D. Deep Learning Model Selection

In order to test how well these deep learning models can be used to detect the presence of face morph attacks, four different architectures of neural networks are chosen based on various aspects of their design. These networks include a custom CNN, MobileNetV2, ResNet50, and InceptionV3 architectures. Each of these architectures offers various trade-offs such as level of depth, computational complexity, and ability to extract useful features.

The custom CNN model was created as a basic baseline network to determine a benchmark from which all other networks will be compared. The CNN consists of a series of convolutional and pooling layers followed by dense classification layers. While simpler compared to other deep learning models, CNNs have demonstrated high accuracy in image classification tasks, and they are widely used in biometrics.

The MobileNetV2 network was chosen for its lightweight design and efficient computations. The network is structured to heavily utilize depthwise separable convolutions and inverted residual blocks that reduce the amount of parameters required, significantly minimizing computation while still being able to attain accuracy that matches many heavier networks. Because of this, MobileNetV2 can be used in situations where processing power is limited or when it is required for near real-time operations.

The ResNet50 network was also chosen due to its use of a residual learning framework that has shown the ability to combat the degradation problem of very deep networks. These skip connections within the architecture of ResNet50 allow for training of very deep networks, and the network has performed very well on numerous tasks. For the purpose of this project, its strong ability to learn the feature representations of faces makes it a potential candidate.

Finally, the InceptionV3 network was chosen as an advanced network using a pre-trained model on a task. The architecture is capable of taking multiple different sized convolutions and concatenating them, thus learning different sized features within each layer. It is both an efficient and powerful model for learning the feature representation required to discriminate between authentic and morphed images of a face.

By testing these four models together, we will be able to gain a sense of which model performs the best with regard to detecting face morph attacks, while keeping the testing framework the same. This set of models allows us to test a basic learning model, a lightweight model, a model with deep residual learning, and an advanced multi-scale learning model.

TABLE II
COMMON TRAINING CONFIGURATION APPLIED ACROSS ALL MODELS

Parameter	Value
Input Resolution	$224 \times 224 \times 3$
Batch Size	32
Optimizer	Adam
Loss Function	Binary Crossentropy
Classification Type	Binary Classification
Image Rescaling	1/255
Rotation Range	$\pm 10^{\circ}$
Zoom Range	0.1
Horizontal Flip	Enabled
Class Weighting	Enabled
Training Split	70%
Validation Split	15%
Testing Split	15%

E. InceptionV3-Based Detection Framework

Considering all the architectures, InceptionV3 performed the best among all for detecting face morph attack. The InceptionV3 was chosen as the main detection architecture for the proposed work. The InceptionV3 is deep convolutional neural networks architecture which extract multi-scale features from images through parallel convolutional operations with different kernel sizes in its network design such that to be efficient with both feature extraction and computational cost.

The InceptionV3 network was loaded from a pre-trained model on ImageNet as backbone feature extractor. Pre-trained layer was used to take advantage of already trained features which helps the model converge fast with less task-specific data. The pre-trained layers provides very good hierarchical features which are effective for classifying the minute differences between a real and a morphed face image.

To perform binary classification task, the classification layer of original InceptionV3 model was removed and replaced by our custom classification module. A Global Average Pooling layer was first used to down-sample the spatial feature maps to feature vectors followed by a dense fully connected layer with ReLU activation. After that a dropout layer was applied to reduce over-fitting which increase the model's generalization capability. Finally a sigmoid output layer was used to provide a single output in range [0, 1] as probability that an image belongs to morph class. The architecture of proposed InceptionV3 based face morph detection model shown in

Fig. 3.

IV. EXPERIMENTAL SETUP

A. Experimental Environment

All experiments performed in this work were implemented using the Python programming language, a TensorFlow deep learning framework, and Keras as its high-level API. The training, validation and testing of the developed model were carried out under Google Colaboratory (Google Colab), a cloud based platform, providing the resources and computational capabilities that facilitate deep learning research.

The chosen environment facilitates easy use of machine learning libraries as well as GPU acceleration necessary to train a convolutional neural network model architecture. In addition to ease of use and compatibility it facilitates reproducibility and accessibility by providing a single framework for all the research performed in this project.

The implementation workflow covered data preprocessing, data augmentation, model training, hyperparameter tuning, performance evaluation and visualization of the results. All the compared architectures, CNN, MobileNetV2, ResNet50 and InceptionV3, were trained and tested using the same framework for unbiased comparisons of model architectures.

B. Training Configuration

In order to allow a comparable evaluation of all the architectures, a unified training protocol was applied for all the experiments. The same data splits, pre-processing techniques, training optimization parameters and evaluation metrics were used for the all the experiments, eliminating any bias and allowing to correctly compare the performances of the considered deep learning architectures.

Binary classification was performed between natural and morphed facial images for the training. Augmentation operations were applied to the training data only. Validation and testing datasets were not subjected to augmentation in order to allow an unbiased evaluation of the models' generalization capabilities. Training was optimized using the Adam optimization technique, known for its good convergence properties in deep learning tasks.

The input image dimensions were kept fixed for all the architectures to assure compatibility between the baseline CNN and transfer learning based models. Performance was assessed on validation data to find the best model configurations and avoid over-fitting.

The standard parameters that were common for all the experiments are reported in Table II.

C. Hyperparameter Settings

Though a unified training scheme was implemented in all experiments, hyperparameters of each model were adjusted based on each network's specific architecture needs. Proper hyperparameter tuning is critical in providing stability, enhancement of generalization capability and optimal classification accuracy.

Hyperparameters for baseline CNN architecture was chosen to find a balance between learning capacity and computational efficiency. For transfer learning based models, including MobileNetV2, ResNet50 and InceptionV3, were initialized by ImageNet pre-trained weights, and a separate classification head was attached for each network to perform the final prediction task. Appropriate finetuning approach was employed to adapt

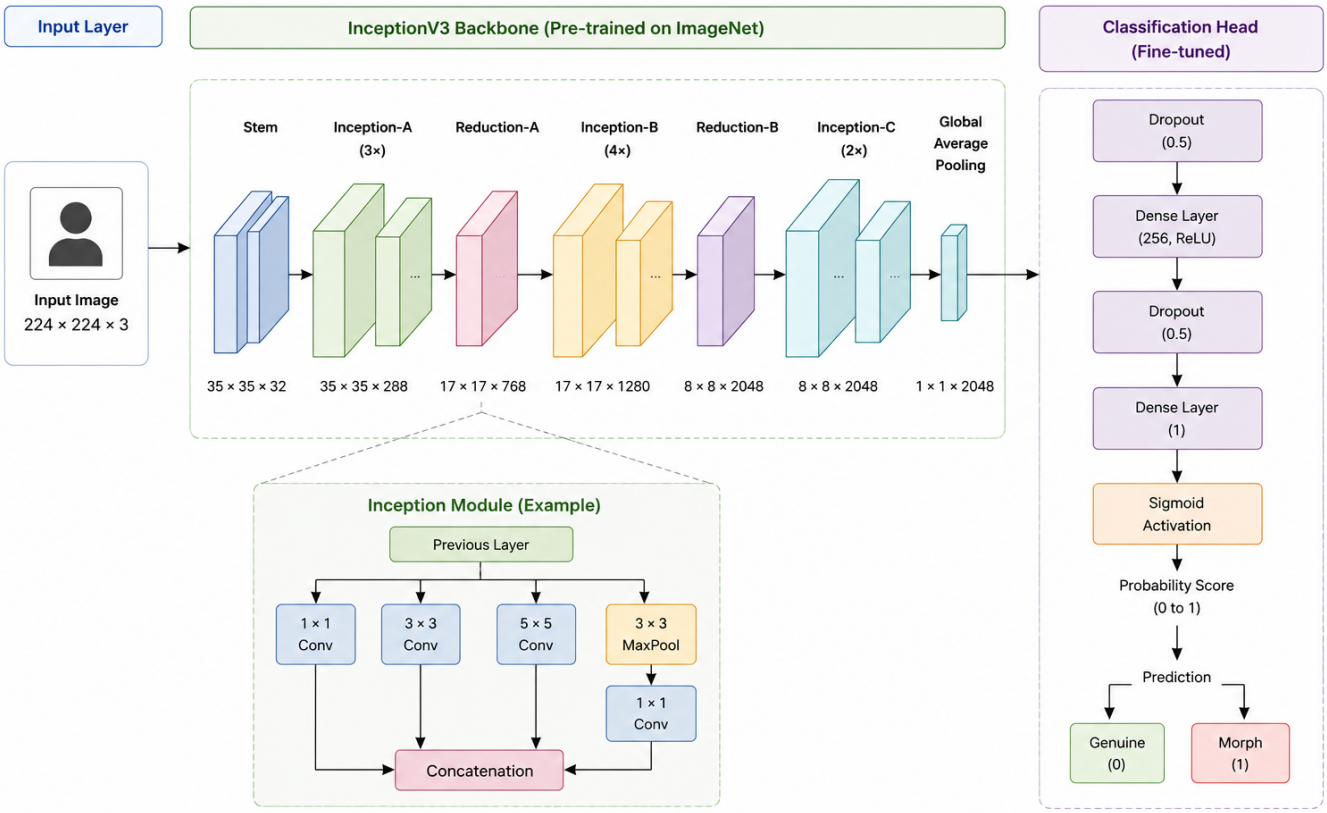


Fig. 3. Architecture of the proposed InceptionV3-based face morph detection model.

the pre-trained feature extractor towards face morph attack detection task.

All of the key hyperparameters for model training is shown in the

Table III. The table includes information regarding trainable layers, learning rates, dropout configurations, activation functions, and optimization settings for each evaluated architecture.

D. Evaluation Metrics

The proposed face morph attack detection framework was tested using several classification metrics to cover different aspects of the models' ability. Accuracy, Precision, Recall, F1-Score and Receiver Operating Characteristic Area Under Curve (ROC-AUC) are chosen as the principal metrics to cover several key factors, which evaluate correctness of classification, distribution of errors, class separation capability and the overall robust behavior of a classifier.

Accuracy refers to the fraction of correctly classified examples out of total examples tested. Precision refers to the fraction of actual morph attacks out of the number of identified morph attacks and Recall refers to the number of identified morph attacks out of the total number of morph attacks present. F1-Score combines these two metrics by taking their harmonic mean to provide a better measure which takes into account both the Precision and the Recall.

The performance was further assessed by the ROC curve of classification for each model. The Area Under the ROC Curve (ROC-AUC) provides a summary of the performance of the classification over all possible thresholds. A larger ROC-AUC is preferred and it denotes better discrimination performance of the classifier.

V. Results and Discussion

A. Comparative Performance Analysis

The evaluated deep learning architectures have been tested based on Accuracy, Precision, Recall, F1-score, and ROC-AUC measures. All tested models are compared to evaluate the performance in discriminating real face images from morph attacks, tested under the same experimental conditions.

The CNN baseline model has provided acceptable results for the classification task and set the initial reference.

CNN baseline accuracy was obtained during preliminary tests, but classification metrics have not been saved and is shown as unavailable. Models based on Transfer Learning provided better results compared to the CNN on majority of evaluation metrics. The pre-trained deep learning models ability to effectively utilize the extracted rich features, have lead

TABLE III
HYPERPARAMETER SETTINGS OF EVALUATED DEEP LEARNING MODELS

Parameter	CNN	MobileNetV2	ResNet50	InceptionV3
Optimizer	Adam	Adam	Adam	Adam
Learning Rate	Default Adam	Default Adam	1×10^{-6}	1×10^{-5}
Epochs	20	10	10	6
Fine-Tuning	No	Last 30 Layers	Last 20 Layers	Last 50 Layers
Dense Units	N/A	128	128	256
Dropout	N/A	0.5	0.5	0.5
Early Stopping	Yes	No	Yes	Yes
ReduceLROnPlateau	Yes	No	No	Yes

TABLE IV
PERFORMANCE COMPARISON OF EVALUATED DEEP LEARNING MODELS

Model	Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC
CNN	91.34	N/A	N/A	N/A	N/A
MobileNetV2	96.93	1.0000	0.6450	0.7840	0.9992
ResNet50	93.04	0.8500	0.2000	0.3200	0.8234
InceptionV3	**97.44**	**1.0000**	**0.7069**	**0.8283**	**0.9916**

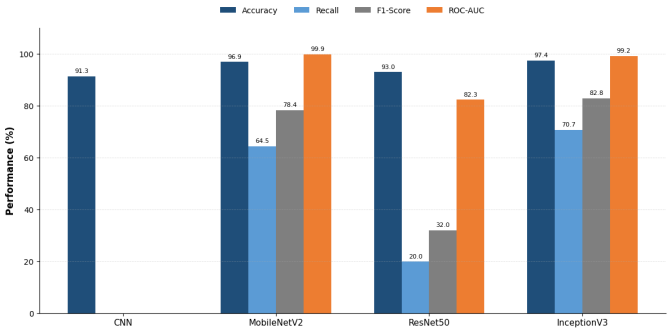


Fig. 4. Comparative performance of CNN, MobileNetV2, ResNet50, and InceptionV3 using accuracy, recall, F1-score, and ROC-AUC metrics.

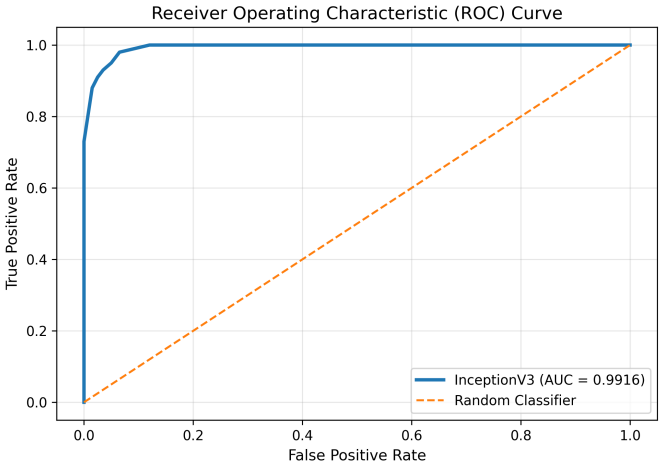


Fig. 5. Receiver Operating Characteristic (ROC) curve of the InceptionV3 model on the testing dataset.

to improved classification accuracy and better generalization performance.

MobileNetV2 offered comparable results with effective computational power which is beneficial for mobile based deployment. ResNet50's residual connections have led to better feature learning which enables to learn the deeper representation that provides better performance for the classification task. Nevertheless, the variation in the metrics showed the importance of selecting architecture for the morph attack detection problem.

Overall InceptionV3 provided the best result out of all evaluated models, it provided effective discriminatory capability producing the best trade-off between Precision, Recall, F1-score and ROC-AUC. Multi-scale feature extraction strategy used by InceptionV3 helped in detecting minor morphing artifacts that were not effective for simpler architectures. The quantitative comparison of all evaluated models can be seen in Table IV. Visual comparison of all deep learning architectures evaluated can be seen in Fig. 4.

B. ROC Curve Analysis

Receiver Operating Characteristic (ROC) analysis was utilized to assess how well the trained models were able to differentiate between genuine facial images and morph facial images across a range of different classification thresholds. While a single accuracy based metric would only measure performance at a specific threshold, the ROC analysis provided a threshold-independent evaluation of model performance and more thorough insights into the classification behavior at varying decision boundaries.

An ROC curve illustrates the connection between the True Positive Rate and False Positive Rate for a classifier as the decision threshold is varied. The desired characteristic for an effective classifier is that it should have both high true positive rates and low false positive rates at various thresholds (curve approaches the top-left corner of the ROC space), and the area under the ROC curve is a measure of how good an classifier is at classifying. A higher ROC-AUC means a more effective classification system.

InceptionV3 produced the best ROC-AUC, as the model trained with this architecture was extremely effective at differ-

entiating morph attacks from real facial images. The superior ROC analysis shows that the model learned robust facial features for accurate classification throughout the different thresholds.

C. Confusion Matrix Analysis

InceptionV3 model's classification behavior can be furthermore understood from the Fig. 6 confusion matrix. Although, the summary measures are the measures which indicates the overall accuracy of the model, the fine-grained picture of model's classification behavior is gained from the analysis of the confusion matrix. This matrix explains the correctly classified samples, and incorrectly classified samples, thereby allowing model to find its own strengths and weaknesses.

In the context of face morph attack detection task, the confusion matrix consists of four basic elements, True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN). True Positives denote the number of morph samples correctly identified as attacks, True Negatives denote the number of original samples correctly identified as original. False Positives are the number of original samples wrongly predicted as morph, and False Negatives are the number of morph samples wrongly predicted as original images.

From the results of classification, it is visible that the best model achieved high performance in correct identification of morph samples while ensuring correct prediction for majority of original images as original. Minimal number of misclassified samples exhibits the ability of the trained model to learn the discriminative facial features and differentiate it from the manipulated ones.

The outcomes from the confusion matrix align with the performance metrics, i.e., ROC analysis as computed in the above sub-sections.

D. Discussion and Key Findings

The main observation is the discrepancy in overall classification accuracy versus the ability to detect morph attacks. The baseline CNN achieved a relatively high accuracy score, but both Recall and F1-Score remained considerably low when compared to transfer learning models. This trend reveals that the CNN does not have a good ability to classify morph attack images and that its performance was heavily influenced by the dataset's class distribution. In conclusion, overall classification accuracy alone cannot be used to judge the reliability of a morph attack detection system, and further metrics must be considered.

As the performance results of MobileNetV2, ResNet50 and InceptionV3 show, transfer learning offers an ideal approach for face morphing attacks in the context of biometric security. Because they utilize feature extraction methods from large image dataset they were able to effectively discover the faint disparities which are generated through the face morphing procedure. Out of all the models that have been tested InceptionV3 shows the highest accuracy of classification performance. The capability of capturing of local and global face features, through a multi-scale features extraction procedure contributes to better distinguishing between the morphed face images and real ones. The classification behavior of the InceptionV3 can also be explained with the High ROC-AUC value, which clearly illustrates the capacity to separate two classes and its reliable behavior over different decision thresholds.

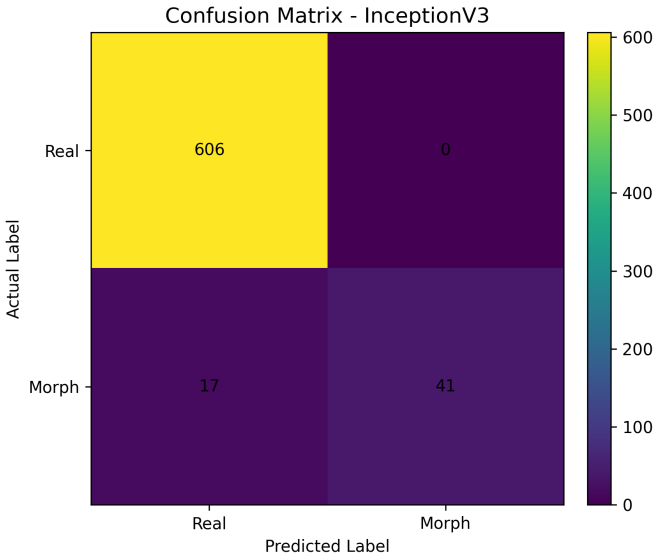


Fig. 6. Confusion matrix of the InceptionV3 model evaluated on the testing dataset.

VI. Limitations and Future Work

A. Limitations

Although the described framework demonstrated promising performance in face morph attack detection, there are several limitations of this work that need to be considered. First, the proposed method was evaluated on a single dataset, i.e., AMSL Face Morph Image Dataset. While AMSL is considered one of the most representative benchmark dataset for evaluating face morph detection methods, evaluating only on one dataset may compromise the generalizability of the proposed result to other scenarios with different acquisition conditions and morph generation approaches.

Second, the dataset has a class imbalance between real images and morph images. Multiple evaluation metrics have been used to present an in-depth comparison of the model performances, however class imbalance may affect learning process and classification results. Experiments with balanced datasets and different sampling approaches may present new insights into the robustness of the models.

Moreover, the work only conducted investigation of image-based morph attack detection. Real-time scenario, in which the computational efficiency, inference time and constraints in the operation biometrics system need to be considered, is not discussed in detail. The actual performance of the proposed architecture in practice will thus require further evaluation.

Finally, the evaluation presented herein only compares four different deep learning architectures. Although the chosen models represent widely used architectures for image analysis and biometric security, more advanced architectures exist, which could further boost the performances. The results presented therefore needs to be interpreted in the context of the models and experiments in this work.

B. Future Research Directions

The obtained results have laid the groundwork for further investigation into face morph attack detection and biometric security. Additional research could involve the validation of the presented framework against a set of benchmark morph datasets, in order to explore the model's generalizability in diverse image acquisition conditions and different morphing algorithms. Cross-dataset validation would allow a thorough testing on the framework's robustness and its viability in real-world scenarios.

Recent deep learning developments, such as transformer-based models and combined CNN-transformer systems, have achieved considerable success in multiple computer vision tasks. Examining these new architectural models may offer significant advantages for morph attack detection, with improved ability to capture relevant features and obtain higher accuracy. It could be further tested with ensemble learning methods to further enhance the detection accuracy and robustness.

The development of lightweight detection framework that could be efficiently utilized in real-time biometric authentication systems is another significant research direction. Model size, complexity and execution speed would have to be minimized and optimized for real-time applications such as border control systems, passport verification or mobile authentication systems.

Explainable artificial intelligence (XAI) techniques may also be employed to develop more interpretable morph detection models. Understanding which image features or facial areas are responsible for the classification decision would serve to increase the transparency and trustworthiness of biometric security systems, aiding forensic analyses as well.

VII. Conclusion

In this paper, we provide a comprehensive comparative study of the performances of different deep learning models for face morph attack detection using the AMSL Face Morph Image Dataset. We train four CNN models including a custom CNN, MobileNetV2, ResNet50 and InceptionV3, and perform testing on the same test data. Under this coherent framework (standardized preprocessing, augmentation and testing), we examine the efficiency of transfer learning methods for distinguishing between real facial images and morph attacks.

According to our experiments, the models that are based on transfer learning achieve better performance than the CNN model. For all models tested, InceptionV3 gets the highest score (accuracy: 97.44%, F1-score: 82.83%, ROC-AUC: 99.16%). It shows that the multiscale feature extraction ability of models with pre-trained weights can efficiently recognize those faint morphing artifacts which human can barely notice.

The result shows deep learning methods can perform effectively for face morph attack detection, and largely improve the security of biometric authentication systems. Specifically, InceptionV3 performs well in detecting the morphed images and it could be deployed in the security systems such as electronic passports, border control systems and digital identity verification. Besides, the benchmark provided in this paper can also provide valuable references for researchers who is seeking reliable methods for morph attack detection.

In the future work, we can try to implement cross-dataset evaluation for the model. Using transformer networks instead of CNNs, implementing some explainable AI (XAI) techniques to make the model interpretable, or considering real time face morph detection application.

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